PROJECT DOCUMENTATIONS

Project Title: SPENDWISE

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Subject:

Java Programming / Full Stack Development

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ABSTRACT

SpendWise is a backend-based expense management system developed using Node.js, Express.js, MongoDB, and JWT authentication. The system allows users to securely register and log in to manage their personal expense transactions.

The application provides REST APIs to create, read, update, and delete transaction records. Authentication is implemented using JSON Web Tokens (JWT) to ensure that only authorized users can access protected routes. MongoDB is used as the database to store user and transaction data.

This project helps in understanding backend development concepts such as API design, authentication, database integration, and CRUD operations.

2.INTRODUCTION

SpendWise is a backend-based expense management application developed using Node.js, Express.js, MongoDB, and JWT authentication. The main purpose of this project is to demonstrate how a secure REST API system works for managing user expenses.

The application allows users to register, log in, and manage their transaction records. Each user can create, view, update, and delete their expenses. JWT authentication is used to secure the API routes and ensure that only authorized users can access their data.

This project helps students understand real-world backend development concepts such as API development, database integration, authentication, and CRUD operations.

3.OBJECTIVE

- The main objectives of the SpendWise project are:
- To develop a backend REST API using Node.js and Express.js
- To implement secure user authentication using JWT
- To perform CRUD operations for managing expenses
- To store and retrieve data using MongoDB database
- To understand backend development architecture and API workflow

4.EXISTING SYSTEM

In the existing system, many people manage their expenses manually using notebooks or simple applications without proper security and data management.

Problems in the existing system:

- No secure authentication system
- Data is not stored in a structured database
- Difficult to manage multiple transactions
- Risk of data loss
- No centralized system to track expenses

5.PROPOSED SYSTEM

The proposed system is SpendWise, which provides a secure and efficient backend solution for managing expenses.

- Features of the proposed system:
- Secure user registration and login
- JWT-based authentication system
- CRUD operations for managing transactions
- MongoDB database for storing data
- Protected API routes for authorized users

This system improves security, efficiency, and data management compared to the existing system.

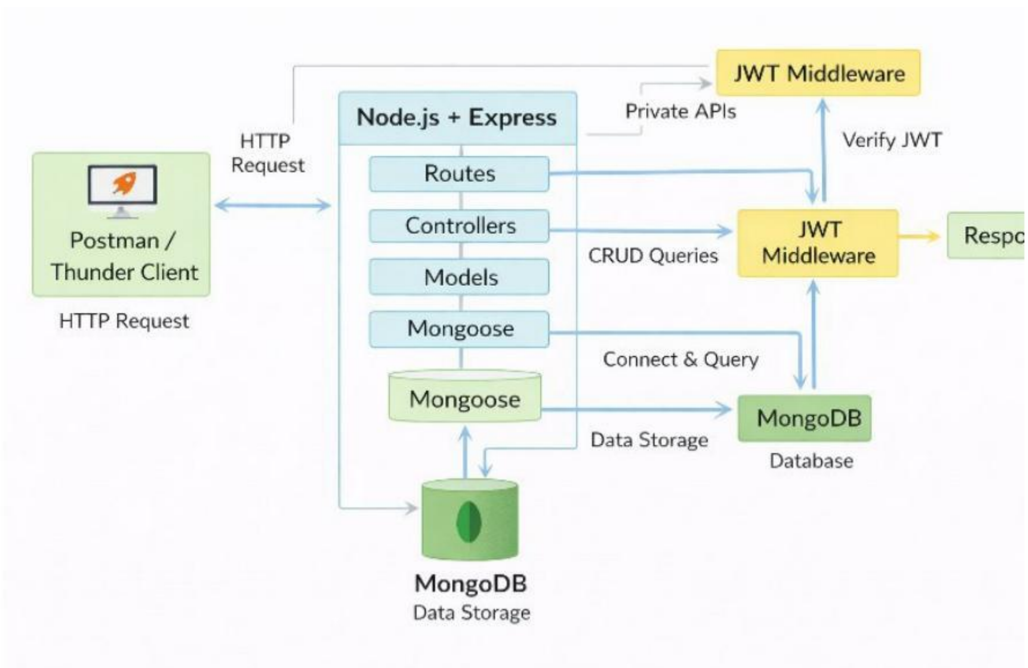
6.TECHNOLOGIES USED

TECHNOLOGY	PURPOSE
Node.js	Runtime environment used to run JavaScript on the server side
Express.js	Web framework used to create REST APIs
MongoDB	NoSQL database used to store user and transaction data
Mongoose	Object Data Modeling (ODM) library for Mongo”B

JWT (JSON Web Token)	Used for user authentication and authorization
Postman / Thunder Client	Used for testing the APIs

7. ARCHITECTURE

The SpendWise project follows a backend REST API architecture consisting of multiple components.



1. Client Layer – The client sends API requests using tools like Postman or Thunder Client.

2. Server Layer (Node.js + Express) – Handles incoming requested and processes business logic.
3. Middleware Layer – JWT middleware verifies the user's authentication token and protects private routes.
4. Controller Layer – Handles CRUD operations for users and transactions.
5. Database Layer (MongoDB) – Stores and manages user and transaction data.

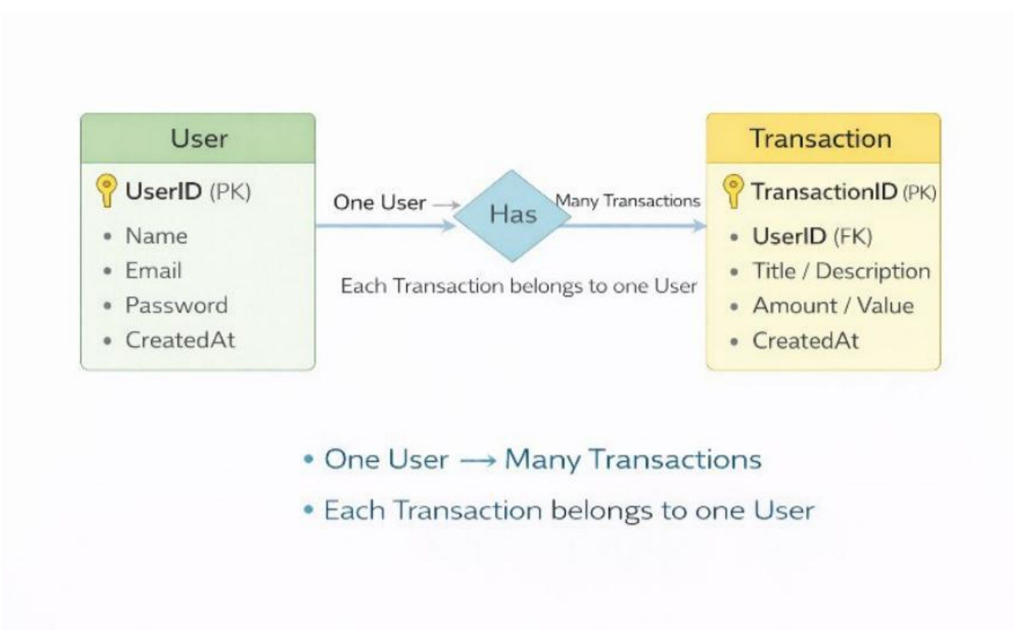
This layered architecture makes the system modular, secure, and easy to maintain.

8.ER DIAGRAM EXPLANATION

The ER Diagram represents the relationship between User and Transaction entities.

User Entity

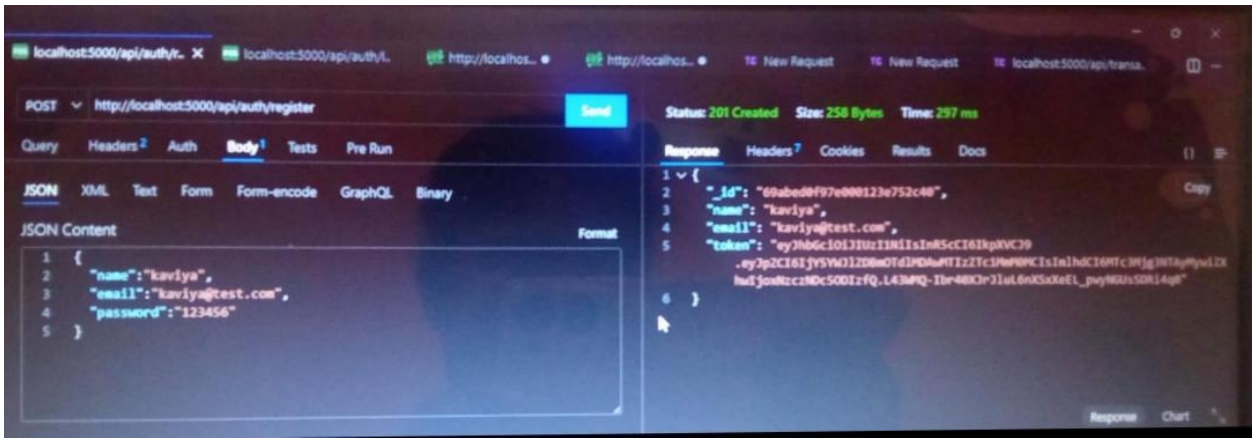
The User entity stores information about the registered users.



Attributes:

- UserID (Primary Key)
- Name
- Email
- Password (hashed)
- CreatedAt

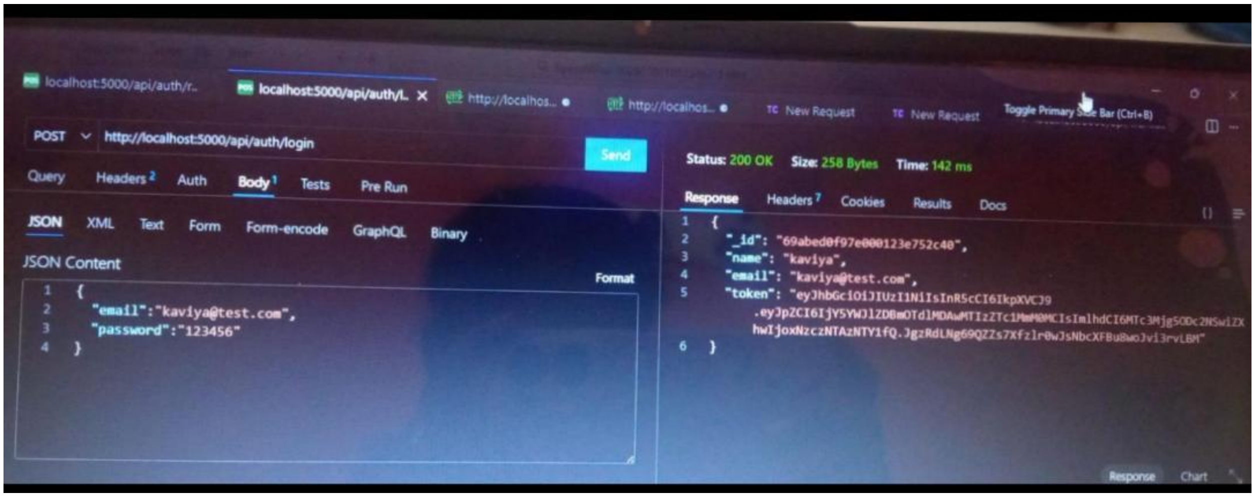
8.1 User Registration Output



Explanation: This output shows the successful registration of a new user in the system.

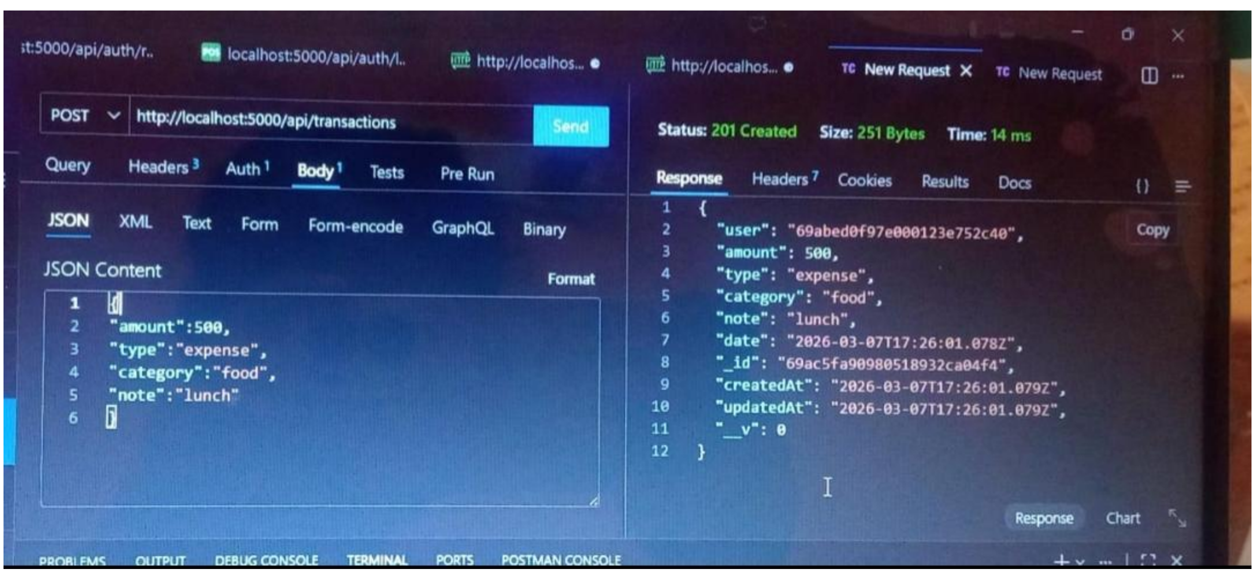
8.2 User Login Output

Explanation: This output demonstrates the login process and generation of JWT



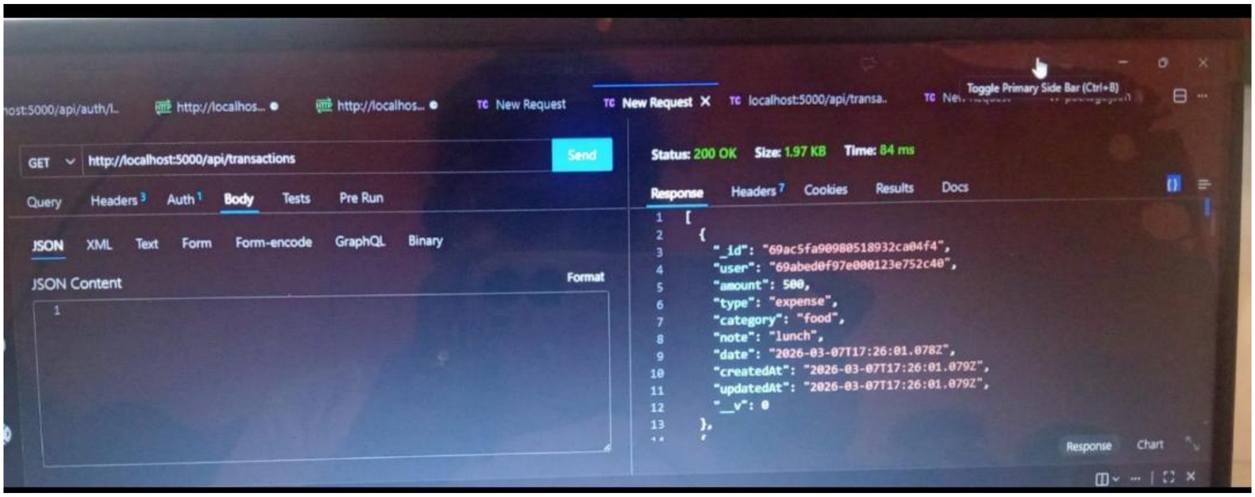
authentication token.

8.3 Create Transaction Output



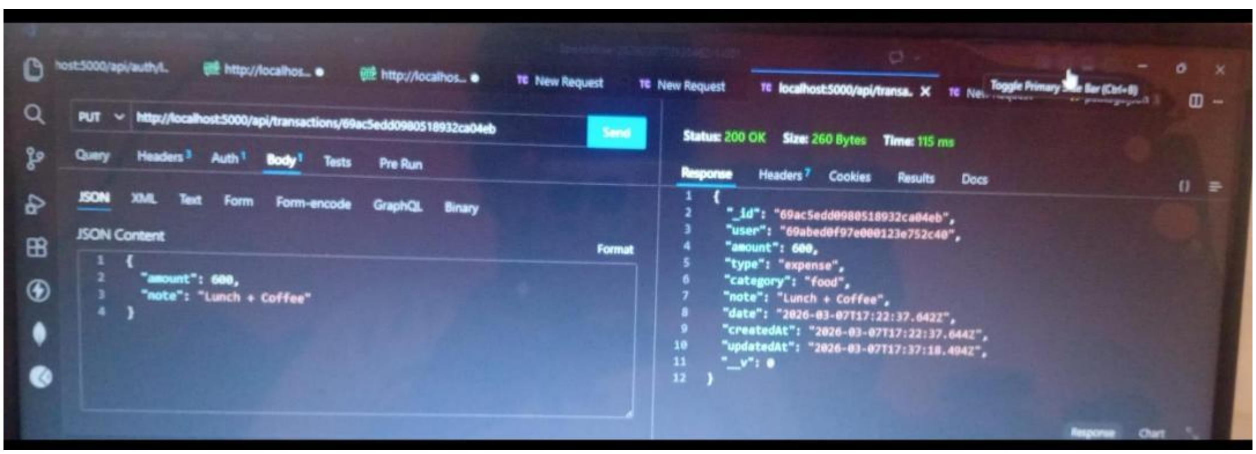
Explanation: This output shows the creation of a new expense transaction stored in MongoDB.

8.4 Get Transaction Output



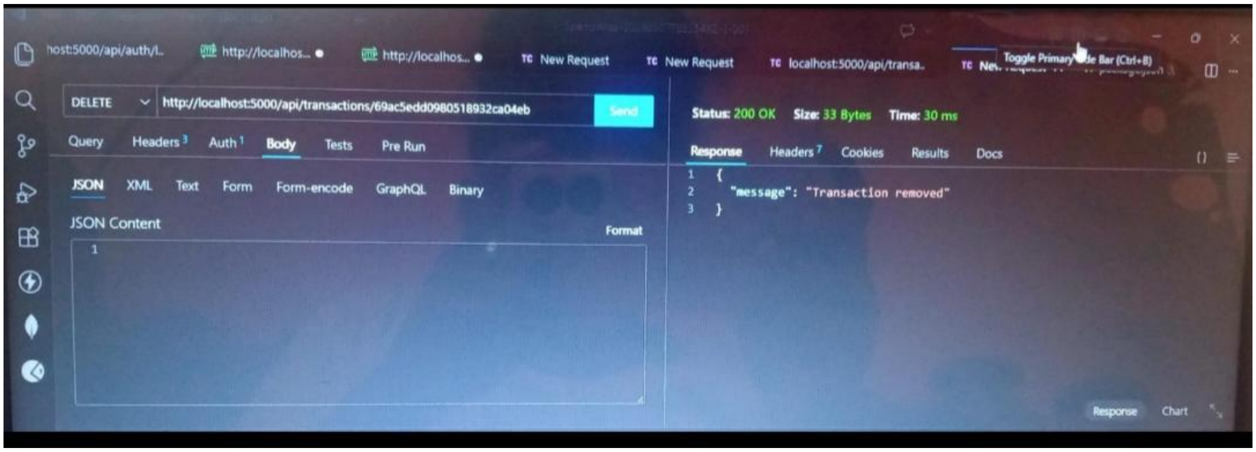
Explanation: This output displays the list of transactions retrieved from the database.

8.5 Update Transaction Output



Explanation: This output shows the successful modification of transaction details.

8.6 Delete Transaction Output



Explanation: This output confirms that the transaction record has been removed from the database

9. FEATURES

- User Registration – Allows new users to create an account.
- User Login – Users can log in securely using email and password.
- JWT Authentication – Generates a secure token for authenticated users.
- CRUD Operations – Users can create, read, update, and delete their transactions.
- Protected Routes – Only authenticated users can access specific APIs.
- MongoDB Integration – Data is stored and retrieved efficiently from the database.

10.ADVANTAGES

- Secure authentication using JWT
- Efficient data storage using MongoDB
- Easy to manage expense records
- Scalable backend architecture

- Helps understand real-world backend development

11.FUTURE ENHANCEMENTS

- Develop a frontend interface using React.js
- Add expense categories and filters
- Generate monthly expense reports
- Implement graphical charts for analysis
- Deploy the system on cloud platforms

12.CONCLUSION

The SpendWise project demonstrates how to build a secure backend application using Node.js, Express.js, MongoDB, and JWT authentication. It provides practical knowledge about REST API development, database management, and user authentication.

This project helps beginners understand how backend systems work in real-world applications and serves as a foundation for developing full-stack applications in the future.

PROJECT DEMO VIDEO

https://drive.google.com/file/d/1rBk5dE4wDs9t06NeLZSJgv0hbkAFuoUq/view?usp=drive_link